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SQL PROJECT ON DELHI METRO ANALYSIS

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PROJECT OVERVIEW:

Analyzed the Delhi Metro network using SQL by exploring station-level and line-level data across multiple metro lines. The project focused on network structure, station distribution, interchange connectivity, and route analysis using advanced SQL queries.

SKILLS DEMONSTRATED:

- SQL Query Writing
- Data Analysis
- Relational Database Concepts
- Data Aggregation
- Join Operations
- Transportation
- Network Analysis
- Problem Solving with SQL

KEY WORK PERFORMED:

- Calculated the total number of stations on each metro line using UNION ALL and aggregate functions.
- Identified junction/interchange stations across multiple metro lines.
- Analyzed and ranked the top 5 longest metro lines based on station count.
- Performed INNER JOINS to discover common stations between metro lines and identify interchange connectivity.
- Used self joins to determine the previous and next station for every station on a metro route.
- Combined data from multiple metro line tables to create a unified network analysis.



Q. How many stations are there on each metro line?

```
SELECT 'Blue Line' AS Line_Name,  
       COUNT(*) AS Total_Stations  
FROM 'blue line'  
  
UNION ALL  
  
SELECT 'Red Line',  
       COUNT(*)  
FROM 'red line'  
  
UNION ALL  
  
SELECT 'Yellow Line',  
       COUNT(*)  
FROM 'yellow line'  
  
UNION ALL  
  
SELECT 'Pink Line',  
       COUNT(*)  
FROM 'pink line'  
  
ORDER BY Total_Stations DESC;
```

Line_Name	Total_Stations
Blue Line	49
Pink Line	37
Yellow Line	36
Red Line	29



Q. Find all junction stations in the Delhi Metro network.

```
SELECT station, 'Blue Line' AS Line
FROM `blue line`
WHERE `is junction` = 1

UNION ALL

SELECT station, 'Blue Line Branch'
FROM `blue line branch`
WHERE `is junction` = 1

UNION ALL

SELECT station, 'Green Line'
FROM `green line`
WHERE `is junction` = 1

UNION ALL

SELECT station, 'Green Line Branch'
FROM `green line branch`
WHERE `is junction` = 1

UNION ALL

SELECT station, 'Red Line'
FROM `red line`
WHERE `is junction` = 1

UNION ALL

SELECT station, 'Yellow Line'
FROM `yellow line`
WHERE `is junction` = 1;
```

station	Line
Dwarka	Blue Line
Janakpuri West	Blue Line
Rajouri Garden	Blue Line
Kirti Nagar	Blue Line
Rajiv Chowk	Blue Line
Mandi House	Blue Line
Yamuna Bank	Blue Line
Mayur Vihar I	Blue Line
Botanical Garden	Blue Line
Noida Sector 52	Blue Line
Karkarduma	Blue Line Branch
Anand Vihar	Blue Line Branch
Ashok Park Main	Green Line Branch
Welcome	Red Line
Kashmere Gate	Red Line
Inderlok	Red Line
Netaji Subhash Place	Red Line
Azadpur	Yellow Line
Kashmere Gate	Yellow Line
New Delhi	Yellow Line
Rajiv Chowk	Yellow Line
Central Secretariat	Yellow Line
INA	Yellow Line
Hauz Khas	Yellow Line
Sikandarpur	Yellow Line



Q. Find all interchange stations in the entire Delhi Metro network

```
SELECT b.Station,
       b.Line AS Line_1,
       y.Line AS Line_2
FROM 'blue line' b
INNER JOIN 'yellow line' y
ON b.Station = y.Station

UNION

SELECT b.Station,
       b.Line,
       m.Line
FROM 'blue line' b
INNER JOIN 'magenta line' m
ON b.Station = m.Station

UNION

SELECT r.Station,
       r.Line,
       y.Line
FROM 'red line' r
INNER JOIN 'yellow line' y
ON r.Station = y.Station;
```

Station	Line_1	Line_2
Rajiv Chowk	Blue Line	Yellow Line
Janakpuri West	Blue Line	Magenta Line
Botanical Garden	Blue Line	Magenta Line
Kashmere Gate	Red Line	Yellow Line

Q. Find previous and next station for every station

```
SELECT cur.Station,  
       prev.Station AS Previous_Station,  
       nxt.Station AS Next_Station  
FROM `blue line` cur  
LEFT JOIN `blue line` prev  
ON cur.`Order` = prev.`Order` + 1  
LEFT JOIN `blue line` nxt  
ON cur.`Order` = nxt.`Order` - 1;
```

Station	Previous_Station	Next_Station
Dwarka Sector 21	NULL	Dwarka Sector 8
Dwarka Sector 8	Dwarka Sector 21	Dwarka Sector 9
Dwarka Sector 9	Dwarka Sector 8	Dwarka Sector 10
Dwarka Sector 10	Dwarka Sector 9	Dwarka Sector 11
Dwarka Sector 11	Dwarka Sector 10	Dwarka Sector 12
Dwarka Sector 12	Dwarka Sector 11	Dwarka Sector 13
Dwarka Sector 13	Dwarka Sector 12	Dwarka Sector 14
Dwarka Sector 14	Dwarka Sector 13	Dwarka
Dwarka	Dwarka Sector 14	Dwarka Mor
Dwarka Mor	Dwarka	Uttam Nagar West
Uttam Nagar West	Dwarka Mor	Uttam Nagar East
Uttam Nagar East	Uttam Nagar West	Janakpuri West
Janakpuri West	Uttam Nagar East	Janakpuri East
Janakpuri East	Janakpuri West	Tilak Nagar
Tilak Nagar	Janakpuri East	Subhash Nagar
Subhash Nagar	Tilak Nagar	Tagore Garden
Tagore Garden	Subhash Nagar	Rajouri Garden
Rajouri Garden	Tagore Garden	Ramesh Nagar
Ramesh Nagar	Rajouri Garden	Moti Nagar
Moti Nagar	Ramesh Nagar	Kirti Nagar
Kirti Nagar	Moti Nagar	Shadipur
Shadipur	Kirti Nagar	Patel Nagar
Patel Nagar	Shadipur	Rajendra Place
Rajendra Place	Patel Nagar	Karol Bagh
Karol Bagh	Rajendra Place	Jhandewalan
Jhandewalan	Karol Bagh	Ramakrishna Ashram Marg
Ramakrishna Ashram Marg	Jhandewalan	Rajiv Chowk
Rajiv Chowk	Ramakrishna Ashram Marg	Barakhambha Road
Barakhambha Road	Rajiv Chowk	Mandi House
Mandi House	Barakhambha Road	Supreme Court

